PLANETARY COUNCIL · POLICY BRIEFING · OCTOBER 2026

Can Britain Feed Itself?

Store and grow: a plan for UK food resilience



Britain grows just enough calories in a normal year and holds no public food reserve. For about **£10 a person a year** it could store food for everyone, and for about **£4.50 more** it could start growing the soil, water, seed and nutrients that every future harvest depends on.

STORE
The UK Strategic Food Reserve
~£10.33 per person per year

GROW
Domestic resilience
~£4.46 per person per year of new money

Every figure carries a numbered reference; the sources for each page are listed at its foot and are clickable. Models, research notes and the full evidence file: github.com/planetarycouncil/food-security-climate-regeneration. **Resilience through transparency: tell us where this breaks.**

OVERVIEW

At a glance

The benefit first: safety and stability for about £10 a person a year.

THE PROBLEM

Britain's farms grow about **101%** of the calories the country needs, a margin of 31 kcal per person per day.[1] That margin rests on imported gas-based fertiliser, imported soya, just-in-time supply chains and less reliable weather. The government has "no national food stockpiles or plans to create these".[2]

PILLAR 1 · STORE

~£10.33 per person a year (£4.93 taxes, £5.40 food prices).[3] Bread for everyone from 4 Mt of food-grade wheat; a 3-day no-cook ration and 90 days of vitamin tablets for all 69.5 million; formula and specialist food; next season's seed and safe nitrogen; ten weeks of soya meal; months of rice, oil and sugar.

Protects against weeks to months: blockades, blackouts, cyber attacks, failed imports.

PILLAR 2 · GROW

~£4.46 per person a year of new money, plus today's £4.87bn of farm support aimed at resilience.[3,4] Soil testing and soil-moisture sensing; farm and catchment reservoirs; UK seed multiplication; nutrient recycling; home-grown protein; support through the switch to regenerative farming.

Protects against seasons to decades: fertiliser and feed dependence, drought, soil loss.

~£14.79

per person a year for both pillars: **under 5p a day**.[3]

WHAT IT IS WORTH

The plan insures a food sector that adds **£162bn** a year to the economy and employs **4.1 million** people, for under 0.5% of that value.[5] It pays for itself if it prevents one 1%-of-GDP shock in 40 years.[3]

ABOVE BOTH PILLARS

A written **contingency ladder**: fair rationing, a livestock destocking order and safe nitrogen, agreed in advance and never improvised.

THE DECISION WORTH TAKING NOW

1. Legislate a statutory food-reserve floor, published release triggers and a power to direct food businesses in a crisis.
2. Ask the ONS how compulsory commercial stocks would be classified, before any purchase.
3. Fund a business case and trials: a mill-by-mill grist trial, a local delivery drill with offline access and water, a dietitian-designed ration.
4. Start the soil-test and soil-moisture expansion, and scale the reservoir grant.

How to read this. Each page builds on the last: the problem and its value (pages 2–3), the evidence (chapters 1–8), the plan (chapters 9–14). The bold line at the foot of each page carries the argument forward.

1. Defra, *Nutritional supply of agriculture in the UK 2025* (15 Oct 2026)
2. UK Parliament, written answer HL15829 (31 Mar 2026)
3. Our cost model, `reserve_basket.py`
4. Defra, *Agriculture in the UK 2025*, ch. 10–11
5. Defra, *Food statistics in your pocket*

What we get: the value of safety and stability

> Safety is invisible when it works. That is why it is underpriced, and why it is worth paying for.

THE TRAGEDY OF THE HORIZON

When a reserve is never used, nothing happens, and nothing happening looks like money wasted. Mark Carney's phrase:[1] the cost of preparing arrives now; the losses it prevents arrive later and land on someone else. Markets value a space company in trillions on hope. Nobody prices what all those companies assume: that the people building them will eat this week.

SMALL BENEFIT, BIG ENOUGH

Instead of promising a large benefit, we show **how small a benefit is enough** for the plan to pay its way. Figures below: store only · both pillars. These are break-even **thresholds**, not forecasts, and they overlap: never add them up.

SAFETY OF THE FOOD SYSTEM

0.44% · 0.63%

of the annual value added by a sector worth £162bn and 4.1 m jobs: about £175 · £250 per food job a year.[2]

PEACE OF MIND

0.0008 · 0.0011

points of average life satisfaction (0–10) pays for it, at HM Treasury's £13,000 per point-year. Too small to measure.[3]

LIVES

~290 · ~410

deaths prevented a year on average would justify it (DfT value of a prevented fatality, £1.8m in 2016 prices, uprated to ~£2.5m: our uprating).[4]

A STABLE ECONOMY

~8 · ~5 years

of the programme cost as much as one week of 10% lower national output (≈ £5.6bn).[5]

SOCIAL STABILITY

Not priced. In the 2000 fuel blockade, supermarkets warned they were days from running out.[6] Rising food prices and unfairness, more than empty fields, are followed by unrest.

REAL, BUT NOT PRICED

Food is harder to use as leverage against a country that holds stock. A buyer that does not have to panic-buy pays less in a crisis. Seed and nitrogen in store protect next year's harvest. The grow pillar rebuilds the soil every harvest stands on.

SCALE

~1p : £1

The UK spends ~£62bn a year on defence, ~£890 per person.[7] The reserve costs about £10: around 1p of food security for every £1 of military security.

So: the plan pays its way at benefits too small to measure. The next eight chapters show why the risk is real.

1. M. Carney, *Breaking the tragedy of the horizon*, Bank of England (2015)
2. Defra, *Food statistics in your pocket* (agri-food GVA £162.3bn, 2024; 4.1 m GB jobs)
3. HM Treasury, *Wellbeing guidance for appraisal* (Green Book supplement)
4. DfT, RAS60: average value of preventing road accidents
5. Our model, `benefits()` in `reserve_basket.py`
6. CNN transcript, UK fuel protests (13 Sep 2000)
7. House of Commons Library, CBP-8175, UK defence spending (2026)

CHAPTER 1

Can Britain feed itself?

Resilient on paper, fragile in practice.

THREE QUESTIONS, NOT ONE



Self-sufficiency, food security and resilience are different questions. This briefing is about **resilience**: what happens in the weeks and months after something breaks.

60%

of the food we eat is produced in the UK, by value (2025); over 80% in the 1980s.[1]

31 kcal

per person per day: the margin between farm output and need (101%), before any waste.[2]

STRONG IN SOME FOODS, WEAK IN OTHERS

By volume Britain is more than self-sufficient in milk, lamb, oats and barley, but grows only **83%** of its wheat, **56%** of its fresh vegetables and **14%** of its fresh fruit (2025).[1]

~1.1 Mt

of protein a year arrives as imported soya meal, mostly fed to pigs and poultry.[3]

ANIMALS AND LAND

Livestock use about **84%** of farmland and supply about **31%** of farm calories.[2,4] The calorie margin would shrink if imported feed stopped, as Defra's own release notes.[2]

PEOPLE AND PRICES

446,200 people work in agriculture; 40% of English farm holders are over 65 and 5% under 35.[5] Food inflation peaked at **19.2%** in March 2023.[6]

0

national food stockpiles, and "no plans to create these" (Defra, March 2026).[7]

READ WITH CARE

The 60% figure is by value and moves with import prices; it is not a calorie share. Defra's calorie statistic assumes no waste. Protein figures differ slightly within Defra's own release (193–195%).

So: Britain grows just enough, but only while gas, soya, lorries and weather all behave. What does it have to work with?

1. Defra, *Agriculture in the UK 2025*, ch. 14: the food chain
2. Defra, *Nutritional supply of agriculture in the UK 2025*
3. AIC soya report (~2.39 Mt meal, 2024) and our arithmetic
4. Defra, *Agriculture in the UK 2025*, ch. 2: structure of the industry
5. Defra, *Farming evidence pack* (Sep 2026)
6. Defra, *Agriculture in the UK 2025*
7. UK Parliament, written answer HL15829 (31 Mar 2026)

CHAPTER 2

Nature’s gift

> World-class farmland per hectare, land-poor per person, with soils that can recover.

MILD, WET, RAIN-FED

Few places grow cereals and grass as well as Britain: a temperate oceanic climate with no dry season, rarely frozen.[1] The lesson is not “grow more of everything”. It is to use the gift wisely: food for people on the best land, grass and trees on the rest.

~7.8 t/ha

typical UK wheat yield, about twice the world average.[2]

17.95 t/ha

world record wheat yield, Lincolnshire, 2022.[3]

0.09 ha

of arable land per person, against ~0.28 ha in France.[8]

WHERE THE BEST LAND IS

About **71%** of UK farmland is grass; England’s best and most versatile land is about **42%** of farmland.[4,9] Much of it lies in the drying east and on shrinking Fen peat, which loses 1–2 cm or more a year under arable. **59.2%** of England’s Grade 1 land lies in mapped river or sea flood-risk areas.[5]

~£1.2bn

a year: the cost of soil degradation in England and Wales (2015 study).[6]

SOILS CAN RECOVER

At Rothamsted, a grass–clover ley rotation raised topsoil carbon from **0.98% to 1.23%** over 28 years, then plateaued; continuous arable declined.[7]



~5 bn L/day

public water shortfall England faces by 2055 without action (the Environment Agency’s do-nothing scenario).[5]

READ WITH CARE

One record field is not the national system. Soil carbon gains are slow, plateau and can be lost. A ley takes land out of crops for years. Mapped flood exposure is not the chance of loss in a given year.

So: the land is good and the soil can heal, but there is little of it per person, and the weather it depends on is changing.

1. Met Office, UK climate zones
2. AHDB, final harvest report (2026); UK average derived from the survey
3. Guinness World Records, highest wheat yield
4. Defra, *Agriculture in the UK 2025*, ch. 2 (table 2.1)
5. Environment Agency, *National framework for water resources 2025*; Grade 1 flood exposure: Environment Agency (2026)
6. Graves et al. 2015, *Ecological Economics* 119
7. Johnston et al. 2017, Rothamsted ley-arable experiment
8. FAO arable land totals ÷ population (our arithmetic)
9. Natural England agricultural land classification survey data

CHAPTER 3

Climate is changing the odds

> British wheat usually rides out a single bad spell. Bad weather stacked across a season is what hurts, and that is becoming more common.

STACKED EXTREMES

What breaks British farming is extremes stacked across one season: wet autumns that stop sowing, dry springs, wet harvests.[1] A reserve is the hedge against **correlated failure** that trade cannot provide.

4 of 5

of England's worst harvests since 1984 came in 2020–2026 (four crops, by volume).[2]

–11%

2026 winter wheat yield against the five-year average (AHDB survey).[3]

1.2%

a year: the chance all major wheat breadbaskets fail together (was 0.3%).[4]

2026, A MIXED HARVEST

Defra's provisional England figures show wheat at **10.8 Mt (+0.8%)**, but barley down 11% and oats at their smallest since 2012. Full UK figures are due in December.[5]

NEIGHBOURS HIT AT ONCE

In 2026 drought hit Britain, France and Germany together. France lost about **8 Mt** of cereal output;[6] EU soft-wheat yields fell about **7%**.[7]

NEW PESTS, WARMER WINTERS

2025 was the UK's warmest year on record.[8] Wheat stem rust is re-emerging[9] and bluetongue is spreading.[10] Wheat self-sufficiency fell from 96% (2023) to 79% (2024) and recovered to 83% (2025).[11]

1,412

water-abstraction restrictions in England by late August 2026.[12]

READ WITH CARE

Average yields may hold up for decades; the risk is in the bad years, not the mean. "Worst harvest" rankings depend on crops, geography and method. Some 2026 European losses reflect less planting because of input costs, not weather alone.

So: bad years are getting more frequent and hit our suppliers at the same time. And a harvest needs far more than weather.

1. Slater et al. 2022, *Earth System Dynamics* 13
2. ECIU, UK harvest even worse than previously thought (Oct 2026)
3. AHDB, final harvest report (2026)
4. Gaupp et al. 2020, *Nature Climate Change*
5. Defra, provisional cereal and oilseed production, England 2026
6. Carbon Brief, Europe's 2026 heat and drought in six charts
7. JRC MARS via UkrAgroConsult (2026)
8. Met Office (2026)
9. EPPO, wheat stem rust in the UK and Ireland
10. GOV.UK, bluetongue
11. Defra, *Agriculture in the UK 2025*, ch. 14
12. Environment Agency abstraction data (27 Aug 2026), via trade press

CHAPTER 4

Hidden dependencies

A loaf of bread runs on gas, diesel, ships and software.

IMPORTED ENERGY, TURNED INTO FOOD

Britain no longer makes ammonia at commercial scale; its last large plant, Billingham, closed in **2023**.[1] All its phosphate is imported, and much of it ends up in sewage.[2] A week-long power cut would break the food system faster than any harvest failure.

~174 kt

of phosphorus imported a year; under half is used productively.[2]

>50%

of the food we eat is chilled or frozen.[3]

CO_2 FOR THE MEAT COUNTER

Food-grade CO_2, used to stun animals and pack meat, has come from a few gas-dependent plants; Billingham once supplied about **60%**.[4]

20 km/h

the speed a Euro VI lorry is forced down to once its AdBlue (made from urea) runs out.[5]

SOFTWARE AND POWER

In May 2025 ransomware at Peter Green Chilled halted orders to most major supermarkets.[6] The Iberian blackout of April 2025 stopped payments, warehouse systems and refrigeration for 50–60 million people.[7]

ONE NARROW SEA

The Short Straits carry about **42%** of dairy and egg imports and **39%** of meat imports.[8]



CRITICAL, BUT UNMAPPED

Food is one of 13 critical national infrastructure sectors, but the NAO says its actual assets (mills, dryers, CO_2, cold stores) still need to be identified.[9]

READ WITH CARE

Phosphorus is a slow risk: soils hold legacy reserves. Manure recycles nutrients already in the system; it is not new supply. Attribution of the 2025 cyber attacks to one group is not established.

So: the weak links are energy, nutrients and logistics, not land. The biggest single one is protein.

1. Defra, *UK Food Security Report 2024*, theme 3
2. Lancaster University, UK Phosphorus Transformation Strategy (2022)
3. Cold Chain Federation, policy strategy 2026
4. Gasworld, UK call for evidence on CO_2 supply resilience
5. Reg. (EU) 582/2011, Annex XIII (retained)
6. Food Ingredients First, ransomware at a UK supermarket supplier (2025)
7. Slimstock, the Iberian blackout and the supply chain
8. Defra, *UK Food Security Report 2024*, theme 3
9. NAO, *Resilience of the food supply chain to disruptions* (Sep 2026)

CHAPTER 5

Protein, livestock and the sea

> Grain isn't the weak point. Imported protein is. And our "animal" grain is a hidden reserve.

~52%

of UK wheat, barley and oats goes to animal feed.[1]

4–6×

more calories go into grain-fed meat than come out.[2]

A HIDDEN RESERVE

Much animal grain could feed people if needed, as porridge, flatbreads and wholegrain, though not as today's sliced loaf without stronger wheat blended in.[3] If imports stopped in a normal year, Britain's grain could keep people and animals going, if industrial uses and exports were cut.[4]

ANIMALS ON THE RIGHT LAND



Keep grass-fed cattle and sheep on land nothing else can use. Only milk and upland suckler beef returned more human-edible protein than they consumed in a 2011 analysis of UK systems.[5] Size pig and poultry numbers to home-grown protein.

WHAT OUR GRAIN MODEL SHOWS

Imports stop on 1 July; distilling and bioethanol cut by 75%; a 1 Mt working floor.[4]

Shock over three years	Extra grain needed
Normal harvests, full livestock feed	2.3 Mt
Two harvests 15% down, feed at 80%	2.5 Mt
A nitrogen-shock harvest 35% down, feed at 80%	3.5 Mt
Three harvests 30% down, feed at 80%	11.4 Mt: storage alone cannot cover it; feed must fall to ~60% even with 4 Mt extra

GROWING THE PROTEIN GAP

Replacing imported soya protein fully would take ~4 Mt of field beans on ~1 Mha, about 16% of cropland. A third within a decade is a realistic aim.[4]

READ WITH CARE

The model pools wheat, barley, oats and maize; a tonne of feed barley is not a loaf. It does not count the feed and CO_2 lost when distilling stops. Cutting UK livestock while importing meat moves emissions abroad.

THE SEA

40%

of seafood eaten is now caught or farmed in the UK (89% in 1975).[6] Contract the winter mackerel and herring catch into cans and pouches.

So: grain is a buffer if we plan to divert it; imported soya is the gap. Both are exposed to the wider world.

1. Defra cereal usage statistics (2023 data)
2. Our arithmetic: 3 kg of grain ≈ 9,900 kcal against ~1,500–2,500 kcal in 1 kg of meat; for global feed context see Mottet et al. 2017, *Global Food Security*
3. UK Flour Millers, annual review
4. Our grain model, `cereal_balance.py`
5. Wilkinson 2011, *Animal* 5(7): re-defining efficiency of feed use by livestock
6. Harrison et al. 2023, *Reviews in Fish Biology and Fisheries*

CHAPTER 6

Geopolitics and prices

> Two chokepoints in one year. Britain buys from neither directly, and pays for both.

2026: HORMUZ AND THE BLACK SEA

The Strait of Hormuz closed[1] and Black Sea grain shipping came under attack.[2] Fertiliser prices doubled within weeks. Britain has felt little so far, because the shock arrives with a lag through next year's fertiliser and harvest.

up to 30%
of traded fertiliser passes through Hormuz.[1]

$400 → $850
per tonne of urea, early 2026 to April.[3]

136.0
FAO Food Price Index, Sep 2026: highest since Nov 2022.[4]

77
food export restrictions by 32 countries in 2022; at the peak ~17% of traded food and feed calories were restricted.[5]

PRICES AT HOME

UK food inflation was **1.3%** in September 2026;[6] spring warnings of 9% have not yet materialised.[7] The UK's carbon border tax starts on imported fertiliser in **January 2027**.[8]

THE RULES

WTO rules allow temporary export restrictions in a food crisis, but exempt World Food Programme humanitarian purchases.[9] The ASEAN+3 rice reserve (787,000 t earmarked) shows neighbours can share stock in calm markets.[10]

A RESERVE IS FOREIGN POLICY

In every big crisis, exporters ban exports and what remains flows to whoever holds stock. A reserve stops Britain panic-buying, lets it keep its own exports open and help partners. In calm years it can earn modest income from storage and rotation sales to aid buyers, at prices agreed in advance, never in a crisis.

READ WITH CARE

Options and swaps fail when export bans bite, so a protected domestic floor comes first. Whether Hormuz is open in late October 2026 is unconfirmed.

So: when supply tightens, stock beats money. And when food gets scarce or dear, people notice who is treated fairly.

1. FAO, chief economist on disruption to the Strait of Hormuz (2026)
2. S&P Global, Black Sea grain trade factbox (Aug 2026)
3. WTO data blog (10 Jul 2026)
4. Fastmarkets, FAO Food Price Index, September 2026
5. IFPRI, food export restrictions tracker
6. ONS, consumer price inflation, September 2026
7. Retail Gazette, IGD warning (Mar 2026)
8. HM Treasury, CBAM factsheet
9. WTO, MC12 decision on WFP food purchases
10. ASEAN, APTERR

CHAPTER 7

Food and stability

> Rising prices, more often than empty fields, are what break the peace. People forgive scarcity sooner than unfairness.

"There are only nine meals between mankind and anarchy."

Alfred Henry Lewis, 1896[1]

LEVELS, NOT SWINGS

Rising food price **levels** are followed by more social unrest across countries; price volatility is not.[2] In rich countries, logistics fail first: in the 2000 fuel blockade, supermarkets warned they would run out within days.[3]

12.2%
of UK households are food-insecure (July 2026).[4]

60% · 4%
of households have three days of no-cook food · three days of water.[5]

2.6 m
emergency food parcels handed out in 2025.[6]

UNFAIRNESS TURNS SHORTAGE INTO DISORDER

Egypt, 1977: cutting the bread subsidy triggered riots in nearly every major city; it was reversed within two days.[7] Japan, 2025: rice prices nearly doubled, released reserve rice reached shops slowly, and the farm minister resigned.[8] WWII rationing gave everyone the same basic entitlement; infant mortality fell during the war.[9]

BRITAIN IS NOT READY

The NAO (September 2026) found local food-disruption planning "insufficient", most resilience forums without a food plan, and no power for Defra to direct food businesses in a catastrophe.[10] In October 2026 over 150 experts wrote to the Prime Minister calling for national food stores.[11]

READ WITH CARE

In Britain the realistic failure is panic buying, empty shelves and lost trust, not mass riots. Unrest has many causes; prices are one.

So: fair, visible preparation is the stability policy. Other countries already do this.

1. Quote Investigator, "nine meals from anarchy"
2. Bellemare 2015, *American Journal of Agricultural Economics* 97(1)
3. CNN transcript, UK fuel protests (13 Sep 2000)
4. Food Foundation, food insecurity tracker (Jul 2026)
5. UK public survey of risk perception and preparedness 2026
6. Trussell, food banks provide over 2.6 million parcels
7. NPR, protests and the price of bread (2011)
8. Food Manufacturing, Japan's rice crisis (2025)
9. Working paper citing Ministry of Health data (arXiv 2301.09982)
10. NAO, summary (Sep 2026)
11. Global Government Forum, experts' letter (Oct 2026)

CHAPTER 8

How other countries prepare

Several comparable countries keep food in reserve. Britain keeps none.

Country	What it holds	How
Finland	At least 6 months of bread grain (about 9 after a 2022–23 top-up)[1]	State agency, funded by a supply-security fee[2]
Switzerland	2–4 months of wheat, sugar, rice, oils, coffee, feed and fertiliser[3]	Compulsory private stocks, rotated through firms' trade
Norway	Rebuilding ~82,500 t of bread grain (~3 months) by 2029[4]	State-owned grain; millers store it on long contracts
Sweden	Grain, seed, fertiliser, feed and spare parts[5]	Law from 1 July 2026 (SFS 2026:837); private custodians
Germany	Federal grain reserve (~1 month of bread) plus rice and pulses[6]	State reserve in ~150 private stores
Japan	Rice reserve, target 1 Mt[7]	State buys, holds ~5 years, sells old rice as feed
Singapore	Two months of rice imports[8]	Licence condition on importers
UK	**None**[9]	

COMPULSORY COMMERCIAL STOCKS WORK

They rotate through normal trade (Switzerland, Singapore). In August 2026 Switzerland released up to **16,000 t** of protein feed because low Rhine and Danube water stalled imports.[10]

RELEASE RULES MATTER AS MUCH AS TONNES

Japan's 2025 release was slow and late, and its refill was criticised as too late.[7] India shows the cost of poor storage: wheat worth ₹700 crore deteriorated in Punjab (2011–16).

READ WITH CARE

Finland's fee and Switzerland's cost (~CHF 14 per person, 2019) cover fuel and medicines too, so they are not like-for-like prices for a UK food reserve.[11] Most national reserve data is incomplete or secret.

So: a reserve is normal policy in comparable countries. Here is what a British one would hold and cost.

1. NESA (Finland), additional grain for emergency stockpiles
2. NESA, funding
3. réservesuisse, compulsory stocks (1 Jan 2026)
4. Norwegian government, emergency grain storage
5. Riksdag, committee report on food stockpiles
6. top agrar, Germany's grain reserves
7. Asia News Network (2025); Adnkronos (Sep 2026)
8. MTI Singapore, reply on stockpiling
9. UK Parliament, written answer HL15829
10. cash.ch, Swiss protein-feed release (Aug 2026)
11. SRF, the Swiss pantry (2019)

STORE: the UK Strategic Food Reserve

About £10 per person per year, under 3p a day, buys bread and a no-cook ration for everyone, the seed and nitrogen for the next harvest, and months of imported staples. Human food comes first, and nobody is left out.

BREAD FIRST



4 Mt of food-grade wheat, held by millers and merchants on top of their normal stock, equals about **ten months** of today's milling with no imports and no harvest, as long as the mills have power, people and the right grist.[1,2] A strong-wheat tranche keeps the loaf rising; a premix keeps flour fortified. About a fifth rotates each year.

NO-COOK, FOR EVERYONE

When the power goes, cookers and bakeries stop. So every person is covered by a **3-day no-cook grain ration** (crispbread, biscuits, oatcakes), stored locally and rotated into schools, the forces and food banks. Vitamin tablets cover everyone for 90 days. On top: infant formula, medical foods, and soft or ready meals for those who need them.[1]

THE NEXT HARVEST

Compulsory stocks hold winter and spring seed, extra pre-season nitrogen held safely as urea and UAN (never solid ammonium nitrate, the Beirut chemical),[3] ten weeks of soya meal, and months of rice, oil and sugar. The state pays for priority fuel, AdBlue and backup power at mills, depots and dryers.

> *Think of it as the National Insurance (not National Insurance).*
>
> Prof. Gen. Adm. F.L. Amingo, Battalion of Militarized Satire

TWO TIERS

A **public crisis tranche** owned by the state and paid from taxes, and **compulsory commercial stocks** (the Swiss model)[4] that firms hold on top of their normal stock and rotate through their own trade, with the cost passed through food prices.

~10 days

of stock held by retailers today;[5] commercial wheat stocks at the pre-harvest low were 1.514 Mt, about 39 days of UK wheat use.[6]

IF MONEY IS SHORT, CUT FROM THE BOTTOM

Cumulative cost per person a year.[1]

Up to	Buys
£2.71	**Bread for everyone:** 4 Mt food-grade wheat, strong wheat, premix, sacks
£3.48	+ **no-cook food for everyone:** 3-day ration, tablets, bottled water
£6.92	+ operations: agency, local plans, priority fuel and power, last mile
£7.55	+ infants and special needs
£9.19	+ next season: seed, safe nitrogen, soya meal
£10.33	+ imported staples: rice, oil, sugar, pulses, oats, milk powder, fish

So: the reserve is built backwards from the meal, for everyone. The numbers follow.

1. Our cost model, `reserve_basket.py`
2. UK Flour Millers, annual review (4.8 Mt milled, 2025)
3. HSE, ammonium nitrate and COMAH
4. réservesuisse, compulsory stocks
5. Full Fact, the UK's food supply
6. AHDB, end-of-season UK cereal supply and demand (Sep 2026)

The reserve in numbers

£718m a year, about 6.8 Mt of stock, and a published answer to every way it could go wrong.

Component	Stock	£ per person a year
Food-grade milling wheat 4 Mt (half in new stores) + 0.35 Mt strong wheat	4.35 Mt	2.62
No-cook grain ration for everyone, 3 days (~500 g a day)	104 kt	0.40
Tablets, flour premix, sacks, 90 m litres of bottled water	–	0.46
Operations: lean agency and inspectorate (~£33m), local plans, priority fuel and power, last mile, security	–	3.44
Infant formula (~7 kt), medical foods, specialist no-cook top-up for 5 m people	–	0.63
Next season: cereal seed, protected urea and UAN-28 (~162 kt N), 10 weeks of soya meal	1.21 Mt	1.64
Imported staples: rice and oil (~4 months), sugar (~2 months), pulses, oats, milk powder, canned fish	1.18 Mt	1.14
Total	**~6.8 Mt**	**~£10.33 (£718m)**

Source: our cost model.[1]

£4.93 · £5.40

via taxes (public tranche) · via food prices (compulsory stocks).[1]

~£3.7bn

of stock and stores to finance up front, public and private, over ~5 years.[1]

Range (sensitivities, not a confidence interval):[1] low £9.40 · central **£10.33** · prices +30% £11.35 · firms financing at 8% £11.92, at 10% £12.73 · grain and fertiliser prices doubled £13.73 · stress £16.56 · severe £18.34.

⚠ WHAT COULD GO WRONG, AND HOW IT IS HANDLED

Risk	Answer
It adds to public debt	Only the ~£0.72bn public tranche is designed to sit on the balance sheet. If the ONS classifies the compulsory stocks to government, debt rises by ~£2.2bn more. Ask first
Firms relabel normal stock	Obligations on top of a registered baseline, checked by an £8m-a-year independent inspectorate
Week one fails without water and delivery	90 m litres covers 10 m people for 3 days; water companies' duties cover the rest and must be drilled. One national food drop would cost ~£400m extra
The same wheat is counted twice	The 4 Mt is the bread floor only; keeping animals fed through two poor harvests needs ~2.5 Mt of separate feed grain[2]
Stored fertiliser explodes	No solid ammonium nitrate: protected urea and heated UAN-28 only[3]
Refill forces bad buying	Refill on a price-and-quality trigger within 36 months; refilling 4 Mt 30% dearer costs ~£0.25bn
It is raided to fight inflation	Release only on published supply triggers, never to manage prices

So: a store buys time, not independence. What we grow with that time is the second pillar.

1. Our cost model, `reserve_basket.py` (central case and sensitivities)
2. Our grain model, `cereal_balance.py`
3. HSE, ammonium nitrate and COMAH

GROW: domestic resilience

A reserve buys time. Soil, water, seed and home-made nutrients decide what we can grow with it.

WHY A SECOND PILLAR

Britain's farms grow about as many calories as the country needs, but every harvest rests on imports and luck.[1] The nitrogen comes from gas made into ammonia abroad; the UK used ~904 kt of fertiliser nitrogen in 2025.[2] A store of food covers a bad season; it cannot cover a decade of dependence.

~£310m

a year of new money: **~£4.46 per person**.[3]

Component	What it buys	£m a year
Soil and water monitoring	~500 soil-moisture sites (from ~50), satellite moisture, a national soil-health baseline, then regular field tests	35
Water storage	Farm and catchment reservoirs that fill in winter: roughly 17,000–50,000 megalitres of new storage a year	100
Seed security	Gene banks, UK seed multiplication (incl. spring cereals and pulses), public breeding	40
Nutrient independence	Manure and digestate testing and precise use; phosphorus recovery; legumes; a green-ammonia pilot	60
Regenerative transition	Independent advice, farmer networks, bridging payments, ~12,000–29,000 ha a year	50
Home-grown protein	Dehulling, pulse mills, local processing	25

Source: our cost model.[3]

WHERE WE START FROM



About **50** soil-moisture stations cover the whole UK today.[4] Reservoir grant rounds have been £10–15m; the August 2026 round was £15m.[5] The UK Vegetable Genebank holds about **14,000** seed samples.[6]

EXISTING MONEY, BETTER AIMED

The UK already spends about **£4.87bn** a year on farm support, roughly £70 per person.[7] The grow pillar asks that it buy resilience: soil, water, protein crops, nutrient efficiency and food on the best land. Retargeting has losers as well as winners, and they should be named.

TARGETS

England targets 40% of farm soils under sustainable management by 2028 and 60% by 2030: management actions, not measured recovery.[8]

READ WITH CARE

£310m a year is a starting envelope and a set of pilots, not national nutrient independence. Regenerative farming often improves annual profit after 3–5 years, but paying back the investment takes nearer nine without support.[9] Pay for measured outcomes, not labels.

So: grow money goes where dependence is deepest. It pays off beyond food too.

1. Defra, *Nutritional supply of agriculture in the UK 2025*
2. Defra, British Survey of Fertiliser Practice 2025
3. Our cost model, `reserve_basket.py`
4. COSMOS-UK soil moisture network, *ESSD* (2021)
5. Defra Farming blog, new drought support (Aug 2026)
6. Farming UK, Warwick seed bank
7. Defra, *Agriculture in the UK 2025*, ch. 10–11
8. AHDB, the revised Environmental Improvement Plan
9. WBCSD, *Closing the gap* (regenerative agriculture in Europe)

Why it matters beyond food



SUPPORTED

Healthier soils at specific sites; erosion control; biodiversity on hedges, margins and flower-rich leys; wellbeing from structured outdoor work.[1,2]

PARTLY SUPPORTED OR CONDITIONAL

Water cycles; lower input costs; climate resilience; cleaner air (only if fertiliser rates fall and slurry is covered); local jobs (more labour per hectare, but net jobs not shown).[3]

WEAK, MIXED OR LIMITED

Nutrient density: weak or mixed. Soil carbon: real but limited; it rises, then plateaus.[1]

VALUES, NOT MEASUREMENTS

Education, community, spiritual connection, future generations.

So: some benefits are proven, some conditional, some are values. We label which is which. Next: who runs all this.

1. Johnston et al. 2017, Rothamsted ley-arable experiment
2. Coventry et al. 2021, nature-based outdoor activity and mental health
3. Organic farming and employment, review

CHAPTER 11

Running it: a lean, open, automated agency

> About 65 people, three deep, heavy on automation, open by default. You cannot hide a thousand jobs and forty warehouses, so we don't try.

LEAN AND THREE DEEP

A **lean agency of about 65 people**, never more than 100 even in a crisis. Every function has a lead and two trained deputies, so leave, illness or resignation never stops anything. Physical inspection is contracted to an independent inspectorate.[1]

MACHINES WATCH, PEOPLE DECIDE

Sensors in every store, automatic samplers at the hubs, and AI watching stock, prices, weather and shipping around the clock. People decide, check, sign and answer to Parliament. A small innovation team keeps improving nutrition, storage, transport and data.

~£33m

a year for the agency, sensors, data and inspectorate.[1]

2 people

must sign every release of public stock.

NEAR PORTS, NEAR PEOPLE



~8–10 port hubs at bulk-grain ports such as Immingham, Tilbury, Liverpool, Avonmouth, Ipswich and Belfast (not Dover, a lorry port). **~30 district stores** across all four nations, much stock inland, away from Humber and Wash flood zones. **Thousands of local points** (schools, councils, depots) holding the no-cook ration. A **statutory minimum in Northern Ireland**.

~1,070

gross jobs across stores, labs, haulage and local planning.[1]

OPEN BY DEFAULT

The agency publishes host towns and ports, monthly regional stocks, buying rules and quarterly volumes, costs, audits and incidents. A short public list of exceptions is withheld: exact fuel, formula and nitrogen sites, live release routes and precise purchase dates. Live dashboard, citizens' panel, annual public stress test and a bounty for anyone who finds a real weakness.

READ WITH CARE

Sixty-five is a commissioning team, to be resized from real workloads; the 24/7 room alone takes about 11 posts. Sensors screen for problems; accredited labs still certify food. Cloud AI can fail first in a blackout, so everything must work on paper and radio, and drills run without the AI.

So: a small, visible team runs a big, boring network. Both pillars also create work worth having.

1. Our cost model and staffing plan, `reserve_basket.py`; research note 23

CHAPTER 12

Work, prestige and a food-producing nation

> Make farming a career people are proud of, let machines do the drudgery, and grow a surplus Britain can sell in good years and eat in bad ones.

TWO TRENDS AT ONCE

AI and automation are hollowing out entry-level office work: graduate adverts are at their lowest since 2016.[1] Meanwhile Britain's farmers are ageing. Both pillars need people: in stores and labs, and as soil testers, water engineers, seed breeders, nutrient advisers and pulse processors.

SCHEMES ALONE DON'T MOVE PEOPLE

In the 2020 Pick for Britain campaign, UK residents filled only about **5–11%** of picking roles.[5] The US Civilian Conservation Corps (1930s) raised participants' health, longevity and lifetime earnings.[6] A work-in-nature pilot (10,000 paid places, ~£350m a year) would be a separate decision.

up to ~3,600

gross jobs across both pillars (not net of displaced work).[2]

−0.64 · −0.94

effect of structured outdoor activity on depression · anxiety (standardised).[3]

A FOOD-PRODUCING NATION

£25.6bn

of food and drink exports in 2025, against £66.9bn of imports.[7] Britain can sell more of what it grows best: oats and barley, dairy, lamb, seed potatoes, breeding expertise. A surplus works as a reserve only if it can be kept at home in a crisis.

REDESIGN THE JOBS

Farm work today is often isolated, seasonal and stressful: **91%** of young farmers call poor mental health the industry's biggest hidden problem.[4] So the jobs need decent pay, training ladders, housing, teams, and machines for the repetitive tasks.

READ WITH CARE

No study yet shows people displaced by AI moving into land work. Prestige follows pay, land access and housing, not slogans.

So: the people side is as real as the grain side. Paying for it fairly needs a new economy of food.

1. Adzuna data via VnExpress (Jul 2026)
2. Our jobs estimate, research note 25
3. Coventry et al. 2021, *SSM – Population Health*
4. Farmers Guardian, Farm Safety Foundation survey (2025)
5. FarmingUK, NFU harvest 2020 survey
6. NBER working paper w27103
7. FDF, trade snapshot 2025

CHAPTER 13

A new economy of food

> Price what harms, pay for what protects, and give the money back.

HIDDEN COSTS

Cheap food has hidden costs: emissions, polluted rivers, degraded soils and diet-related disease. Carbon pricing is reaching farming anyway, starting with a border tax on imported fertiliser in January 2027.[1]

~12%

of UK greenhouse gas emissions come from agriculture, including 49% of methane and 59% of nitrous oxide (2024).[2]

≥ $10 trillion

a year: the hidden costs of the world's food system.[3]

£0.4–1bn

a year: Danish-style livestock tax rates applied to UK livestock, all recycled (our estimate).[4]

OUR POSITION

One consistent carbon price across jurisdictions: no free allowances, no subsidies for emitting, and CO_2 treated as a commodity. Emitters pay; verified removers earn. Link the UK and EU schemes and phase out free allowances jointly;[5] price avoided emissions, temporary storage and durable removals as different products; phase out fossil breaks such as the red-diesel rebate gradually, returning the money to the same farms.

DENMARK'S MODEL

Tax livestock emissions from 2030, deduct most of it, and recycle the revenue into farming.[4] The lesson of Europe's 2024 farmer protests: change imposed on farmers fails, change that pays them works.[6]

CREDITS THAT DON'T HOLD

Soil-carbon credits have been shown to track rainfall more than management in Australia.[7] Biodiversity net gain statutory credits start at **£42,000** each, two per unit: a deliberately expensive last resort.[8] UK farm support was £4.87bn in 2025, including a red-diesel rebate of about £384m.[2]

READ WITH CARE

Full carbon pricing raises food prices, so a per-person dividend matters. Pricing UK livestock without a border adjustment on food imports exports the emissions. Do not spend the same revenue twice.

So: prices that tell the truth fund the transition. Here is the plan, step by step.

1. Farmers Weekly, the carbon tax on fertiliser
2. Defra, *Agriculture in the UK 2025*, ch. 10–11
3. FAO, *The State of Food and Agriculture 2023*
4. The Local Denmark, livestock CO_2 tax (2024); UK range: our estimate
5. ICAP, UK Emissions Trading Scheme
6. Euronews, EU withdraws pesticide law (Feb 2024)
7. Mitchell et al. 2024, *Carbon Management*
8. GOV.UK, statutory biodiversity credit prices

CHAPTER 14

The plan

> Store and grow. Plan the next step. Fewer, better animals. Close the loops. Price the truth. Bring people with us.

WHEN

When	Store	Grow
0–12 months	Legislate the reserve, a statutory minimum, release triggers and a power to direct; ask the ONS; buy food-grade grain only in good harvests; local food plans in every resilience forum	Start the soil-test and soil-moisture expansion; scale the reservoir grant; aim farm support at soil, water, protein and nutrient efficiency
1–3 years	Phase in compulsory stocks; build the public tranche and the universal no-cook ration; agree and exercise the destocking order and rationing rules	Seed multiplication and public breeding; first reservoirs filling; manure testing; phosphorus recovery; pulse processing
3–10 years	Mature, rotating, audited reserve in all four nations	Soya imports down by about a third; national soil-health map; green-ammonia pilot

THE CONTINGENCY LADDER

Level	Situation	What happens
0 · Normal	Business as usual	The reserve rotates; monthly public stock report; exercises
1 · Disruption	Days to weeks: fuel blockade, cyber attack, blackout	No-cook food, priority fuel and power for mills and depots, purchase limits
2 · Sustained shortage	Months: import cut, input shock, one failed harvest	Staged release; feed-to-food switching; distilling cut; destocking stage 1
3 · Severe	Two poor harvests plus fertiliser shortage	**Fair Shares Guarantee**: the same basic entitlement for everyone, top-ups by need, capped staple prices, paper fallback, automatic exit
4 · Catastrophic	Volcanic winter, war, global crop disease	All edible grain to people; international mutual aid

THE LIVESTOCK DESTOCKING ORDER

Written in advance with farmers and vets: stop new births first; simplify processing to bulk cuts for public catering; keep lawful by-product routes open; pre-agreed compensation; and never let animals starve.

THE LAW

Emergency regulations under the Civil Contingencies Act must be approved within 7 days and lapse after 30.[1] A Food Security (Emergency) Act should build the standing scheme; the emergency powers switch it on. The **Cabinet Office holds the release key; Defra operates**, with all four governments at the table.

So: store and grow, triggered by written rules, not panic. One related idea sits outside the costed plan.

1. Civil Contingencies Act 2004, Part 2 (ss. 26–27)

RELATED CONCEPT · NOT PART OF THE COSTED PLAN

Homes, Land and Livelihoods

> Build homes where the work is. New settlements can give a new generation a home, land to work and a reason to stay.

BRITAIN HAS DONE THIS BEFORE

The garden cities of Letchworth (1903) and Welwyn (1920), the post-war new towns, Milton Keynes (1967), Cambourne in Cambridgeshire (from 1998),[1] and now **Salt Cross** garden village near Eynsham, whose plan for about 2,200 homes and a science park was adopted in February 2026.[2] Since COVID, many people no longer have to live near an office.

THE IDEA



Modern, efficient homes **clustered** near an existing village or town, with **shared** farmland held by a community land trust, a machinery ring, a pack house and a cold store, so residents can work in food, nature and land for some or all of the week. The modern "mule" is shared kit, skills and working capital, not a plot each.

DESIGN RULES

- **The home never depends on the farm job.** If a farm business fails, residents keep their homes.
- **Affordability comes from tenure**, through a land trust or housing association, not a "smallholding" label.
- **Keep the best land in food:** build on the edge of existing places or on previously developed land, never on peat or flood plain.
- **Count whole-life carbon**, including car journeys.
- **Pilot two or three sites first** against ordinary affordable housing plus a separate farm-entry scheme.

1.62 ha

(four acres) per household would take ~9.6% of UK farmland for a million homes.[3]

~£500k

of sales a shared farm needs to pay four people; the village alone buys ~£52k.[3]

READ WITH CARE

More homes help affordability locally, but national prices are driven mostly by interest rates.[4] Planning is slow and contested. The phrase "forty acres and a mule" is not used: it was a broken promise of land to formerly enslaved Americans.

So: a pilot worth testing, not a promise. The last page lists what still needs checking.

1. South Cambridgeshire District Council, new communities
2. West Oxfordshire DC, Salt Cross AAP adoption statement (Feb 2026); Salt Cross FAQs
3. Our land arithmetic, research note 26
4. Miles & Monro 2019, Bank of England staff working paper

OPEN MATERIALS

Resilience through transparency

> We publish our own critique. Tell us where this breaks.

HOW IT WAS CHECKED

This briefing has been through four rounds of independent adversarial review. In the full evidence file every claim carries a label: **[P]** primary source, **[S]** secondary, **[E]** our estimate, **[I]** interpretation or position, **[?]** not yet verified.

STILL TO CHECK BEFORE ANYONE SPENDS MONEY

1. How the ONS would classify compulsory commercial stocks, and what firms would really pay to finance them.
2. Whether mills can run emergency grists, plant by plant, with backup power.
3. A dietitian-designed ration with variants for infants, pregnancy, coeliac disease, allergies and people who cannot chew.
4. A priced last-mile and water service, tested in a real local drill.
5. Quotes for storage, new stores, reservoirs and fuel contracts.
6. Full UK 2026 harvest figures (due December 2026) and Defra's reply to our FOI request.[1]

OPEN MATERIALS

- The full evidence file: every claim, source and counter-argument.[2]
- Two Python models: the cost of both pillars and a monthly grain-stock model.[3]
- 27 research notes.[4]

An army defends the border. A food reserve defends the dinner table.

1. WhatDoTheyKnow, FOI "Strategic food reserves and compound risk" (Oct 2026)
2. Full evidence file, `draft/briefing_full_context.md`
3. Models, `research/model/`
4. Research notes, `research/notes/`